



ELE335 (Artificial Intelligence)

INTEGRATED MEDICAL EXPERT SYSTEM

Presented by:

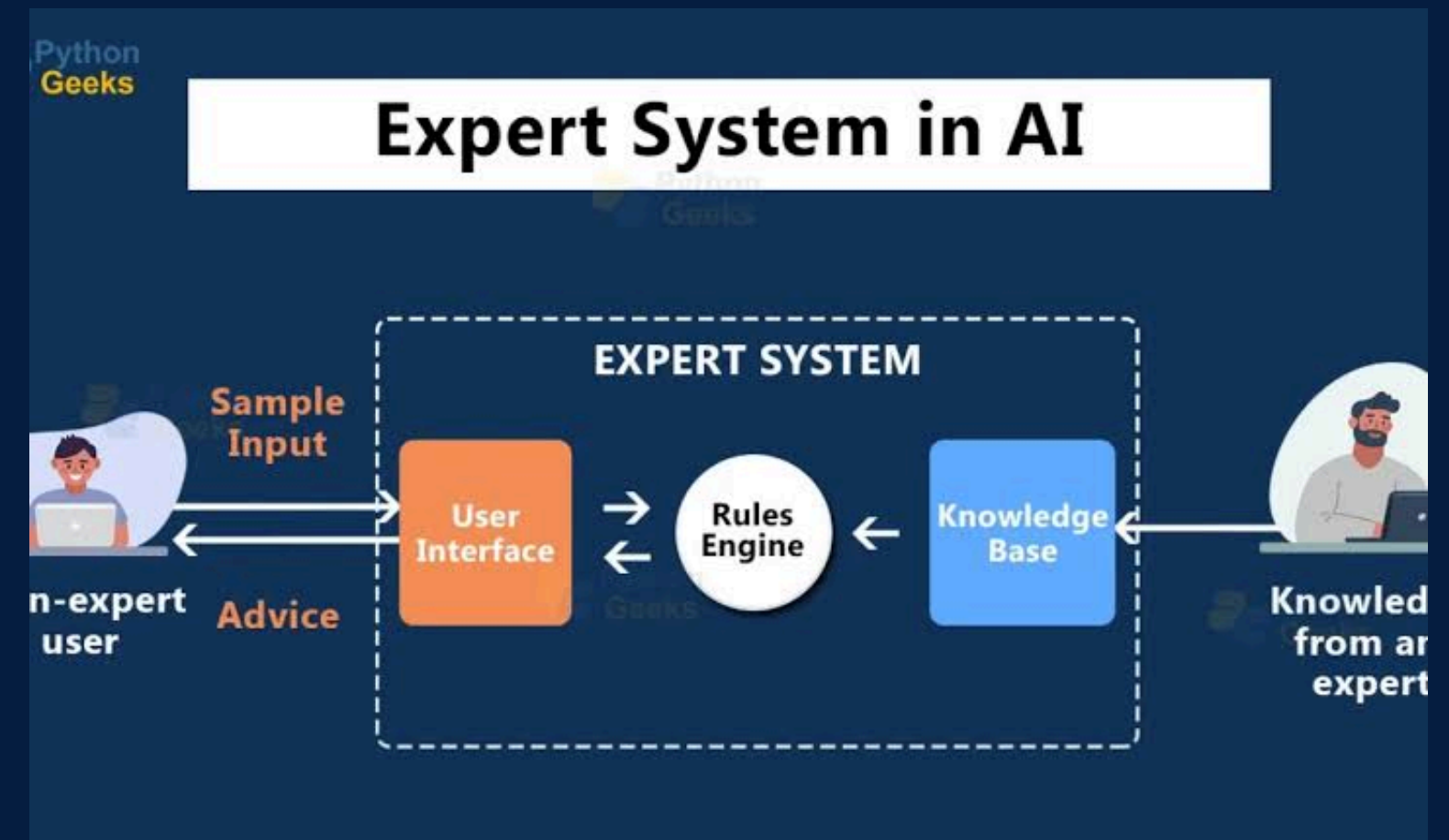
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INTRODUCTION

This project introduces an expert medical diagnosis system developed to assist users in identifying potential diseases based on the symptoms they experience. By answering a set of Yes/No questions, the system can determine the most likely condition and provide appropriate treatment advice. The system was built using Prolog, which is well-suited for logic-based tasks and expert systems. It supports both Arabic and English to be accessible for a wider range of users.



PROJECT'S FEATURES

- **Symptom-based Diagnosis:**

Asks users a series of Yes/No questions and identifies the most probable disease based on their answers.

- **Treatment Recommendations:**

Provides clear and simple treatment advice after diagnosis to guide the user on next steps.

- **Bilingual Support:**

The system supports both Arabic and English to reach a wider range of users.

- **Expert System Logic:**

Built using Prolog, which allows easy updates to the medical knowledge base by adding or modifying rules and facts.

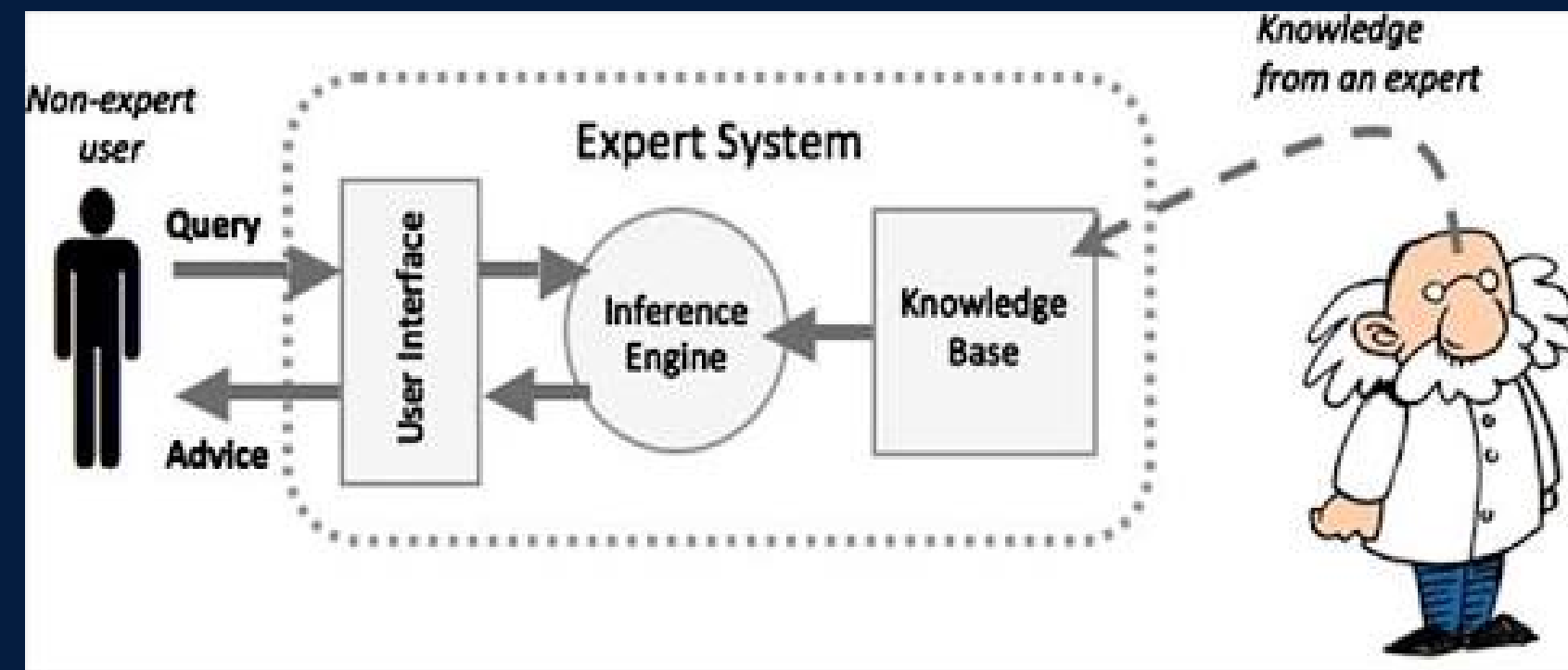
- **Expandable Design:**

The system is designed in a way that allows easy integration into future mobile or desktop applications.

OVERFLOW OF PROJECT

The system follows a simple yet effective process:

- The user is asked a series of Yes/No questions about their symptoms.
- The system checks these responses against a database of rules that represent different diseases.
- Once a match is found, the system displays the disease name and its suggested treatment.
- If no match is found, it informs the user that a diagnosis couldn't be made and recommends consulting a real doctor.



SAMPLE OUTPUT

```
2 ?- start.  
Choose language (en/ar): en  
-----  
          Integrated Medical Diagnosis System  
-----  
Important Notice: This system is for educational purposes only.  
It is not a substitute for professional medical diagnosis.  
Suggested treatments are for general guidance. Please consult a doctor before use.  
  
Enter patient name (or type "stop" to exit): aya  
  
Please answer with yes/no to each symptom.
```


SAMPLE OUTPUT

```
- common_cold: 25.0%  
  (Low confidence. Do not treat without professional advice)  
- sinusitis: 50.0%  
  Suggested treatment:~n  * Steroid nasal sprays  
  * Oral decongestants  
  * Saline nasal wash  
- pneumonia: 60.0%  
  Suggested treatment:~n  * Amoxicillin-clavulanate (625mg every 8 hours)  
  * Complete rest  
  * Oxygen if saturation < 92%  
- flu: 66.7%  
  Suggested treatment:~n  * Rest at home  
  * Warm fluids  
  * Paracetamol (500â€“1000mg every 6 hours)  
  * Aspirin (if not allergic)
```



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THANKS, THE END

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